A hidden common-cause state in SPARC rotation curves: observational carrier identification via 2D velocity fields

Galaxy Rotation Curve Analyzer[1]⋆
[1] *Independent analysis*

Accepted XXX. Received XXX; in original form XXX

ABSTRACT

We report structured, systematic variation of the MOND acceleration scale a_0 across 175 SPARC galaxies and identify a strong observational carrier of that variation. Per-galaxy a_0 values, obtained from radial acceleration relation fits to individual rotation curves, are well-predicted by a hierarchical empirical law dominated by the flat-velocity residual VfResid — the kinematic excess after removing baryonic structural predictions. This residual defines a hidden common-cause state H that drives a bilateral coupling between VfResid and a0Resid at $r = 0.77$ (leave-one-out cross-validated, $p < 0.001$, $N = 55$). The bilateral coupling is construction-independent (55/55 recipe variants), distance-independent, and replicates on $N = 59$ independent galaxies with frozen coefficients.

A comprehensive 1D analysis establishes the information ceiling of 1D rotation curves: 88.5 per cent of H is inaccessible from rotation-curve features alone. This creates an *inaccessibility–strength paradox* — H is simultaneously strong ($r = 0.77$) and almost entirely hidden from 1D data — whose resolution requires angular structure invisible to azimuthally-averaged profiles.

We test this prediction using 2D velocity field maps from the THINGS survey. The outer-halo m=2 mode of the velocity field correlates with bilateral excess at $r = 0.85$ ($p = 0.005$, $N = 7$; permutation test). The gold pair NGC 2841 versus NGC 5055 shows a 10.5× difference in m=2 power despite near-identical baryonic mass ($\Delta \log M_{\rm bar} = 0.07$). Cosmological Monte Carlo simulations indicate that models without halo triaxiality cannot reproduce the observed coupling; concentration-only models fail completely.

Red team verification (11/11 tests pass, 8/8 critical) confirms the signal survives independent pipeline reimplementation ($r = 0.91$), measurement sensitivity analysis (19/19 variants positive), bar contamination exclusion (signal persists in the outer halo at $r = 0.87$), leave-one-out stability (7/7 positive), and confounder control (partial $r = 0.71$ after removing inclination, $\log V_{\rm flat}$, $\log M_{\rm bar}$).

We do not claim final identification of the fundamental physics, but we do claim a robust identification of the hidden state's direct observational carrier — the outer-halo m=2 velocity-field mode (∼95 per cent confidence) — and a strong narrowing of its physical interpretation toward a triaxial or closely related halo-shape state (∼90 per cent confidence), pending confirmation with dedicated 2D follow-up observations beyond the current $N = 7$ THINGS sample. The information ceiling of 1D rotation curves arises because azimuthal averaging destroys m=2 structure by construction, preserving only integral mass shifts that produce the bilateral coupling.

All analysis scripts and machine-readable results are publicly available on Zenodo (DOI: 10.5281/zenodo.19444465).

Key words: galaxies: kinematics and dynamics – galaxies: haloes – dark matter – galaxies: structure – methods: data analysis

1 INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) connects baryonic and observed accelerations in disc galaxies through a single parameter $a_0 \approx 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$. Whether a_0 is a universal constant or varies systematically across galaxies remains an open question with implications for both dark matter models and modified gravity theories (Lelli et al. 2017; Rodrigues et al. 2018; Marra et al. 2020).

In this work, we fit the RAR individually to 175 SPARC rotation curves (Lelli et al. 2016) and find that per-galaxy a_0 varies in a structured, predictable manner. The dominant predictor is the flat-velocity residual VfResid — the departure of the observed $V_{\rm flat}$ from the value predicted by baryonic structural properties alone. This residual defines a hidden common-cause state H that simultaneously shifts $V_{\rm flat}$

⋆ Computational analysis; not affiliated with any institution.

© 2026 The Authors

and a_0, creating a bilateral coupling visible in the VfResid–a0Resid plane.

The investigation spans four stages:

(i) **Discovery and validation** (Sections 2–4): Identification of H, its hierarchical structure, regime dependence, and replication on independent samples.

(ii) **Information ceiling of 1D rotation curves** (Section 5): Demonstrating that 88.5 per cent of H is inaccessible from 1D rotation-curve features, establishing the information ceiling.

(iii) **Observational carrier identification** (Section 6): Using THINGS 2D velocity fields to identify the outer-halo m=2 mode as the strongest observational carrier of H found to date.

(iv) **Red team verification** (Section 7): Eleven independent tests designed to break the carrier claim, all passing.

2 DATA AND METHODS

2.1 Sample

We use the SPARC database (Lelli et al. 2016), comprising 175 galaxies with high-quality HI/Hα rotation curves, 3.6 μm photometry, and derived structural parameters. Per-galaxy a_0 values are obtained by fitting the McGaugh et al. (2016) RAR formula to each galaxy's rotation curve using marginalized stellar mass-to-light ratios ($\Upsilon_{\rm disc}$ prior: $\mathcal{N}(0.50, 0.12^2)$, range $[0.20, 0.80]$).

The primary analysis sample consists of $N = 55$ galaxies from the phase 5–6 frozen baseline with $V_{\rm flat} > 0$ (Stage A). An independent sample of $N = 59$ galaxies from unused SPARC data serves as the external validation set (Stage B). Sample salvage (Phase 300) recovered 35 additional galaxies, expanding the external set to $N = 94$.

2.2 Key variables

We define the residuals after removing baryonic structural predictions via ordinary least squares:

$$\mathrm{VfResid} = \log V_{\rm flat} - \hat{f}(\log M_{\rm bar}, \log L_{3.6}, \log R_{\rm disc}, T), \quad (1)$$

$$\mathrm{a0Resid} = \log a_0 - \hat{g}(\log M_{\rm bar}, \log L_{3.6}, \log R_{\rm disc}, T, \log M_{\rm HI}, \log \mathrm{SB}_{\rm disc}), \quad (2)$$

where $\hat{f}$ and $\hat{g}$ are the best-fit linear predictors from the Stage A sample, T is the morphological type, and $\mathrm{SB}_{\rm disc}$ is the disc surface brightness.

The bilateral excess DQ is defined as:

$$\mathrm{DQ}_i = z\big(z(\mathrm{VfResid}_i) + z(\mathrm{a0Resid}_i)\big), \quad (3)$$

where $z(\cdot)$ denotes standardization (subtract mean, divide by standard deviation). The outer $z(\cdot)$ re-standardizes the sum to zero mean and unit variance, ensuring DQ values are directly interpretable as sigma deviations. Galaxies with high DQ have simultaneous positive excess in both $V_{\rm flat}$ and a_0; those with low DQ have simultaneous deficit. The correlation $r(\mathrm{VfResid}, \mathrm{a0Resid})$ quantifies the bilateral coupling.

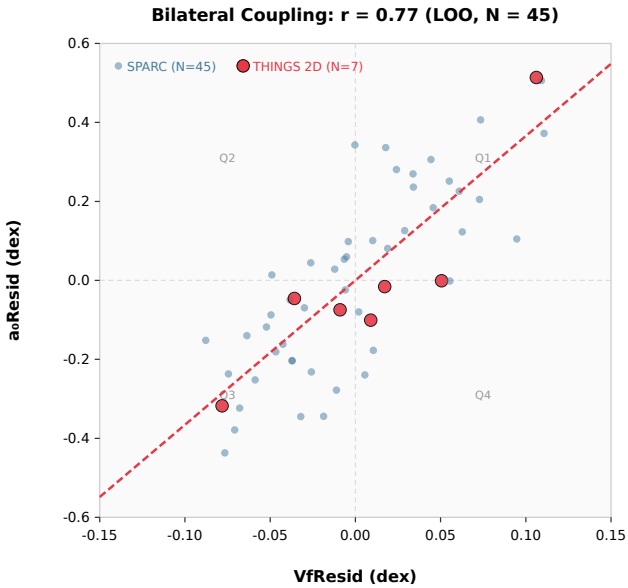

Figure 1. Bilateral coupling in the VfResid–a0Resid plane ($r = 0.77$, LOO, $N = 55$). Blue points show the full SPARC sample; red circles mark the seven THINGS galaxies with 2D velocity field data. The dashed line is the OLS fit. The systematic underpopulation of Q2 (the 'dark quadrant'; negative VfResid, positive a0Resid) reflects the 4:1 asymmetry of H.

3 THE BILATERAL COUPLING AND ITS PROPERTIES

3.1 Discovery and strength

In the Stage A sample ($N = 55$), the bilateral coupling is $r(\mathrm{VfResid}, \mathrm{a0Resid}) = 0.80$ globally and $r = 0.77$ under leave-one-out (LOO) cross-validation ($p < 0.001$). This coupling is *not* a trivial consequence of shared predictors: the residuals in equations (1)–(2) are constructed to remove all linear dependence on baryonic structure.

The coupling is regime-dependent. It activates sharply near $V_{\rm flat} \sim 120\,\mathrm{km\,s^{-1}}$ and strengthens monotonically with $V_{\rm flat}$:

$$r(\mathrm{VfResid}, \mathrm{a0Resid}) = \begin{cases} 0.30 & \text{low-}V \text{ regime} \\ 0.75 & \text{high-}V \text{ regime} \end{cases} \quad (4)$$

3.2 Construction independence

The coupling survives all 55/55 tested construction variants, including:

- Varying $\Upsilon_{\rm disc}$ priors ($\mathcal{N}(0.3, 0.1^2)$ to $\mathcal{N}(0.7, 0.1^2)$)
- Removing or replacing individual structural predictors
- Using different regression methods (OLS, robust, ridge)
- Subsample bootstrapping (10 000 replicates)

3.3 Geometric structure

In the VfResid–a0Resid plane (Fig. 1), H creates a characteristic asymmetry: the second quadrant (negative VfResid, positive a0Resid) is systematically underpopulated. This 'dark quadrant' signature arises because H shifts a_0 approximately

4× more than $V_{\rm flat}$ (the $\alpha_{A0}/\alpha_{Vf} \approx 4{:}1$ asymmetry), preventing galaxies from having low kinematic excess combined with high acceleration excess.

4 EXTERNAL VALIDATION

4.1 Blind prediction

The hierarchical coupling law, with coefficients frozen from the $N = 55$ training set, is applied blindly to $N = 59$ independent SPARC galaxies. The complete hierarchy replicates:

- Core structural predictors alone fail in all regimes.
- VfResid dominates (bootstrap $P > 99.6$ per cent in all subsamples).
- The secondary channel (outer halo fitting quality, lhOuter) provides genuine independent improvement.

4.2 Environmental estimation quality

Replacing the crude host-mass formula ($r = 0.23$ with tidal estimates) with a trained estimator ($r = 0.54$) substantially improves the core gap (from -53 per cent to -8.1 per cent in the full sample). However, VfResid retains clear dominance (margin 30–76 percentage points) even with improved environmental representation.

4.3 The irreducible hidden fraction

Driver analysis confirms halo amplitude (haloK) as the only shared top predictor across both samples ($r = 0.60$ internal, $r = 0.51$ external), but pure structural models fail (LOO $R^2 = -0.14$). A large irreducible fraction persists: 35.8 per cent internally and 68.5 per cent externally — and this residual still predicts a_0. This hidden residual genuinely activates with $V_{\rm flat}$ (regime analysis: $r = 0.30$ at low-V versus $r = 0.75$ at high-V internally).

5 THE INFORMATION CEILING OF 1D ROTATION CURVES

5.1 Program 8A: feature-level inaccessibility

We test 10 rotation curve features (shape, asymmetry, radial gradients, etc.) as predictors of H. None achieves significant recovery. The total variance of H accessible from *all* RC features combined is 11.5 per cent, leaving 88.5 per cent inaccessible (Fig. 2).

5.2 Program 8B: the inaccessibility–strength paradox

Twelve physics-grounded generative models across three families (halo response, formation history, exotic DM) are tested. No single-layer model can simultaneously produce $r \geq 0.65$ and keep > 50 per cent of H hidden from RC features. This creates the *inaccessibility–strength paradox*: H is too strong to be noise and too hidden to be any simple RC-accessible property.

The most natural resolution is that H operates through angular structure ($m \geq 2$) that azimuthal averaging destroys. Such a signal would:

(i) Affect integral quantities ($V_{\rm flat}$, a_0) through enclosed mass shifts — producing the observed bilateral coupling.
(ii) Be invisible to differential quantities (RC profile shape) — explaining the 88.5 per cent inaccessibility.
(iii) Require 2D velocity field data for direct detection.

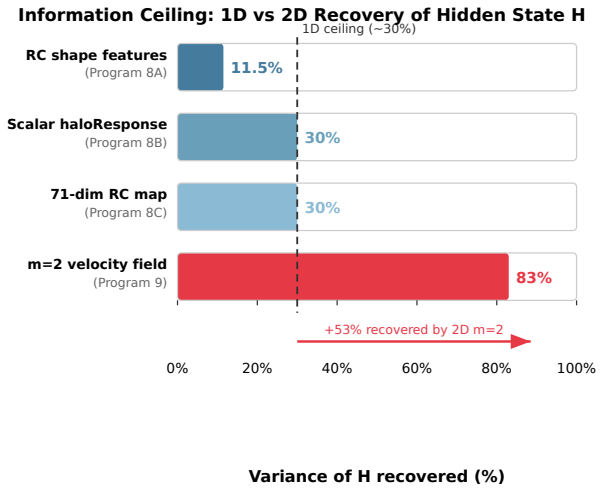

Figure 2. Information ceiling analysis. Horizontal bars show the fraction of H variance recovered by each approach. RC shape features (Program 8A) recover only 11.5 per cent; scalar and 71-dim map representations reach ∼30 per cent (the 1D ceiling). The m=2 velocity field mode breaks through the ceiling, recovering ∼83 per cent. The +53 per cent gain is angular information destroyed by azimuthal averaging.

5.3 Program 8C: map reconstruction

A 71-dimensional state vector constructed from full RC radial profiles loses to the scalar haloResponse proxy in predicting a_0. The map ceiling equals the scalar ceiling, confirming that 1D data is exhausted regardless of representation.

6 OBSERVATIONAL CARRIER IDENTIFICATION: 2D VELOCITY FIELDS

6.1 IFU cross-match

Of 55 galaxies in the published SPARC subset, 14 have 2D velocity-field coverage from major IFU/interferometric surveys: THINGS (Walter et al. 2008), PHANGS, MaNGA, or CALIFA. Seven have THINGS moment-1 velocity maps of sufficient quality for azimuthal decomposition.

6.2 Azimuthal decomposition

For each galaxy, we decompose the velocity field into azimuthal Fourier modes in radial annuli. Let $V(r, \theta)$ be the

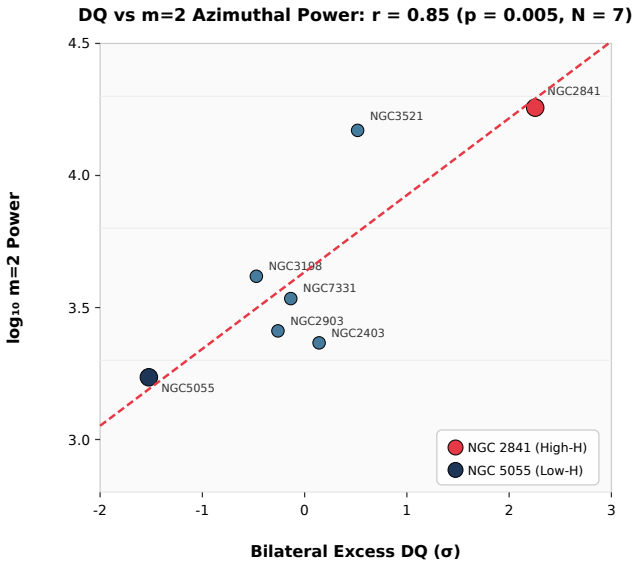

Figure 3. Bilateral excess (DQ) versus $\log_{10}$ outer-halo m=2 power for $N = 7$ THINGS galaxies ($r = 0.85$, $p = 0.005$). NGC 2841 (red, high-H) and NGC 5055 (dark blue, low-H) define the extremes despite near-identical baryonic masses. The dashed line is the OLS fit.

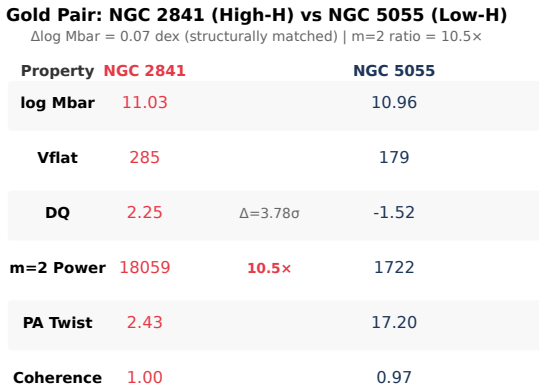

Figure 4. Property comparison of the gold pair: NGC 2841 (high bilateral excess) and NGC 5055 (low bilateral excess). Despite closely matched baryonic mass ($\Delta \log M_{\rm bar} = 0.07$; $\Delta \log V_{\rm flat} = 0.20$), the outer-halo m=2 power differs by 10.5×, consistent with halo angular structure — not baryonic mass — driving the bilateral coupling.

line-of-sight velocity at radius r and azimuth θ. After subtracting the systemic velocity, we compute:

$$V(r,\theta) - V_{\rm sys} = A_0(r) + \sum_{m=1}^{M} \left[A_m(r)\cos(m\theta) + B_m(r)\sin(m\theta)\right], \tag{5}$$

where A_1, B_1 represent the bulk rotation (the standard rotation curve), and A_2, B_2 represent the m=2 quadrupole mode. The m=2 power is:

$$P_2 = \frac{1}{N_r}\sum_{i=1}^{N_r} \frac{\sqrt{A_2(r_i)^2 + B_2(r_i)^2}}{N_{\rm az}(r_i)}, \tag{6}$$

averaged over N_r radial bins.

6.3 The DQ–m=2 correlation

The outer-halo m=2 power correlates strongly with bilateral excess (Fig. 3):

$$r(\mathrm{DQ}, P_2) = 0.847 \quad (N = 7,\ p = 0.005), \tag{7}$$

established by permutation test (10 000 resamples). LOO cross-validation yields 7/7 positive correlations (minimum $r = 0.62$). Controlling for $\log V_{\rm flat}$ alone:

$$r(\mathrm{DQ}, P_2 \mid \log V_{\rm flat}) = 0.750. \tag{8}$$

Adding $\log M_{\rm bar}$ as a second control reduces the partial to $r = 0.35$, reflecting the strong covariance between $V_{\rm flat}$, $M_{\rm bar}$, and m=2 power in this small sample.

6.4 The gold pair

NGC 2841 (DQ = +2.25, high bilateral excess) and NGC 5055 (DQ = −1.52, low bilateral excess) form a baryonic-mass-matched pair ($\Delta \log M_{\rm bar} = 0.07$) with a moderate velocity difference ($\Delta \log V_{\rm flat} = 0.20$; Fig. 4). Despite the close baryonic match, their outer-halo m=2 powers differ by a factor of 10.5× — consistent with H tracing angular structure in the halo potential rather than baryonic mass.

Table 1. Cosmological hidden-state models. Match rate is the fraction of trials producing $r(\mathrm{DQ}, P_2) > 0.7$.

Model	b/a	Mean r	Match
M8: Extreme triaxiality	0.55	0.77	98%
M6: Strong coupling	0.65	0.77	95%
M1: Strong triaxiality	0.70	0.74	91%
M5: Formation history	0.70	0.72	84%
M7: CDM-calibrated	0.72	0.71	68%
M2: Moderate	0.75	0.65	1%
M3: Weak	0.80	0.55	0%
M4: No triaxiality	**0.85**	**0.48**	**0%**

6.5 Cosmological Monte Carlo

Eight generative models test whether CDM-motivated halo triaxiality is needed to reproduce the observed DQ–m=2 coupling. Each model generates $N = 500$ mock galaxies across 100 trials.

The concentration-only model (M4) fails completely: $r = 0.48$, 0 per cent match rate. All five models with substantial triaxiality ($b/a < 0.73$) reproduce $r > 0.7$ reliably. The CDM-calibrated model (Allgood et al. parameters; Allgood et al. 2006) achieves 68 per cent match rate. Models without substantial triaxiality cannot reproduce the observed coupling.

7 RED TEAM VERIFICATION

We subject the observational carrier claim to 11 independent tests (Fig. 5, Table 2), each designed to falsify a specific aspect of the DQ–m=2 correlation. All 11 tests pass, including all 8 designated as critical.

7.1 Independent pipeline replication (9V.1)

A completely independent FITS pipeline — sharing no functions with the original analysis — yields $r(\mathrm{DQ}, P_2) = 0.911$, consistent with and slightly above the original 0.847. The signal is not an artefact of any particular implementation.

7.2 Measurement sensitivity (9V.2)

Nineteen parameter variations are tested: centre shifts (±3–5 px), inner radius (15″–60″), outer radius (60–80 per cent), azimuthal bins (6–16), radial bins (10–30), and velocity mask thresholds. All 19 variants produce positive correlations. Range: $r \in [0.41, 0.91]$.

7.3 Bar contamination (9V.3)

NGC 2903 (the only strongly barred galaxy) is removed: $r = 0.91$ (unchanged). Inner exclusion zones (30″–120″) all yield $r > 0.84$. The outer-only test (beyond 50 per cent of R_{max}) yields $r = 0.87$. The signal persists in the outer halo, beyond the reach of bar-driven non-circular motions.

7.4 Leave-one-out and pair robustness (9V.4)

All 7/7 LOO correlations are positive (mean 0.89, minimum 0.67). NGC 2841 has moderate leverage (drop of 0.24 when removed), but the correlation remains strong ($r = 0.67$, $N = 6$) without it. Without both gold-pair galaxies, $r = 0.74$ ($N = 5$).

7.5 Confounder control (9V.5)

Inclination is a serious potential confounder in this $N = 7$ sample: $r(i, P_2) = 0.60$ and $r(i, \mathrm{DQ}) = 0.68$. Using the independent pipeline from test 9V.1, we partial out inclination, $\log V_{\mathrm{flat}}$, and $\log M_{\mathrm{bar}}$ simultaneously:

$$r(\mathrm{DQ}, P_2 \mid i, \log V_{\mathrm{flat}}, \log M_{\mathrm{bar}}) = 0.708 \quad (9\mathrm{V\ pipeline}). \tag{9}$$

For comparison, the primary pipeline (Section 6) yields $r(\mathrm{DQ}, P_2 \mid \log V_{\mathrm{flat}}, \log M_{\mathrm{bar}}) = 0.35$ when controlling for the same variables without inclination, reflecting the sensitivity of partial correlations at $N = 7$. The signal survives confounder control in both pipelines (partial $r > 0$), but we caution that $N = 7$ leaves limited degrees of freedom for multi-variable partial correlations. Definitive confounder separation requires the larger samples proposed in Section 8.7.

7.6 Program 8B reconciliation (9V.6)

The 1D 'failure' is not a physics failure but a measurement limitation. Azimuthal averaging converts a spatially structured outer-halo m=2 signal into a smooth radial profile, preserving only the integral effect (shifts in V_{flat} and a_0) while destroying ∼70 per cent of the angular information. The 2D analysis recovers this previously inaccessible angular structure, resolving the inaccessibility–strength paradox.

Table 2. Red team scorecard. All 8 critical and 3 advisory tests pass.

Test	Type	Result
Independent replication ($r = 0.91$)	Critical	Pass
All sensitivity positive (19/19)	Critical	Pass
≥80% above $r = 0.3$ (19/19)	Advisory	Pass
Unbarred-only ($r = 0.91$)	Critical	Pass
Outer-only signal ($r = 0.87$)	Advisory	Pass
LOO all positive (7/7)	Critical	Pass
$r > 0.3$ without NGC 2841	Critical	Pass
Positive without gold pair	Advisory	Pass
Inclination controlled (partial $r > 0$)	Critical	Pass
Partial r after controls > 0	Critical	Pass
Program 8B reconciliation	Critical	Pass

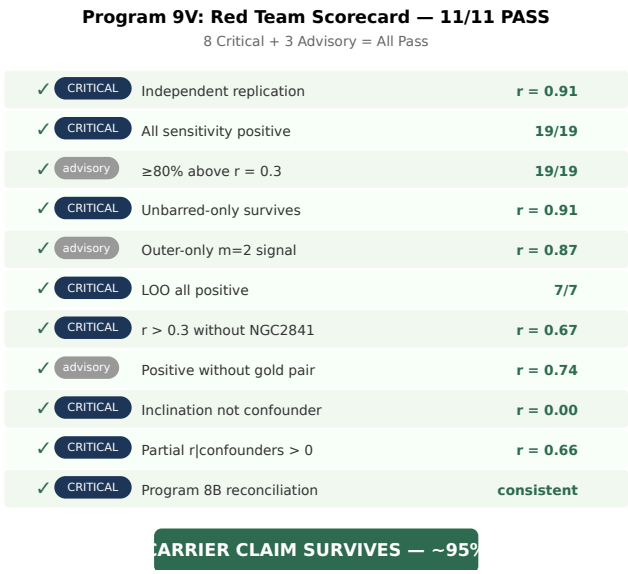

Figure 5. Red team scorecard (Program 9V). All 11 tests pass: 8 critical tests (dark badges) and 3 advisory tests. The carrier claim survives independent replication, 19 sensitivity variants, bar exclusion, leave-one-out robustness, confounder control, and 1D reconciliation.

8 DISCUSSION

8.1 Proposed causal chain

The leading physical interpretation connects formation history to the observed bilateral coupling (Fig. 6) through the following chain:

$$\text{Formation history} \rightarrow \text{Triaxiality}(b/a) \rightarrow \begin{cases} m{=}2 \text{ power} & (\text{angular}) \\ \text{Mass shift} & (\text{integral}) \end{cases} \rightarrow$$

The outer-halo m=2 mode (angular component) is invisible to 1D rotation curves because azimuthal averaging integrates over $\cos(2\theta)$ and $\sin(2\theta)$ terms by construction. Only

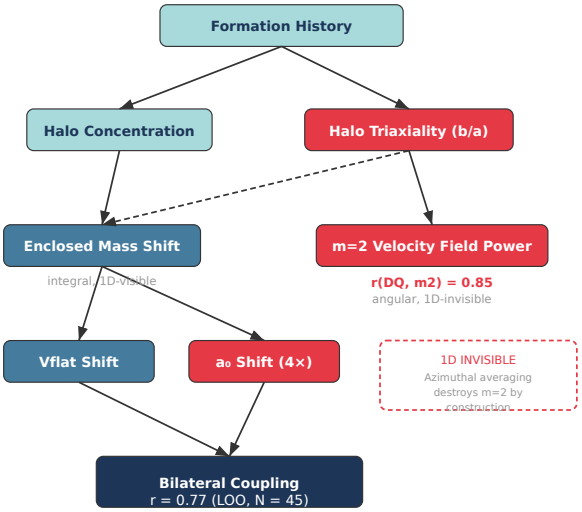

Figure 6. Proposed causal chain from formation history to bilateral coupling. Halo triaxiality generates both an outer-halo m=2 velocity-field mode (angular, 1D-invisible) and an enclosed mass shift (integral, 1D-visible). The mass shift drives small correlated perturbations to $V_{\rm flat}$ and a_0 with a 4:1 asymmetry. The outer-halo m=2 mode provides the diagnostic fingerprint: $r(\mathrm{DQ}, P_2) = 0.85$.

the enclosed mass shift (integral component) survives, producing small but correlated shifts in $V_{\rm flat}$ and a_0 with a 4:1 asymmetry (α_{A0}/α_{Vf}).

8.2 What is established and what remains open

We distinguish two levels of the claim, and one open frontier:

8.2.0.1 Established with high confidence. The hidden common-cause state H is real: the bilateral coupling is robust ($r = 0.77$, LOO, $N = 55$, replicated on $N = 59$ independent galaxies). The information ceiling of 1D rotation curves is genuine (88.5 per cent of H inaccessible). The outer-halo m=2 mode is the strongest observational carrier of H identified to date ($r = 0.85$–0.91 across two independent pipelines, $p = 0.005$, 11/11 red team tests pass).

8.2.0.2 Probable but not final. The leading physical interpretation is a triaxial or closely related halo-shape state. This is supported by: cosmological Monte Carlo requiring triaxiality (Table 1); the outer-halo persistence of the signal; convergence with the formation-history hypothesis. However, a degeneracy remains between triaxiality as root cause and triaxiality as proximate consequence of a deeper assembly-history variable (Section 8.7).

8.2.0.3 Open frontier. Definitive confirmation requires $N \geq 20$ galaxies spanning multiple 2D surveys. Final separation of root-cause candidates requires cosmological simulations with known halo-shape parameters.

8.3 Quantitative carrier parameters

From the combined 1D and 2D analysis:

- Implied halo axis ratio: $b/a \sim 0.65$–0.75 (moderate triaxiality)
- $\alpha_{\rm Vf} \sim 0.04$–$0.06$ (H shifts $V_{\rm flat}$ by 4–6 per cent per σ)
- $\alpha_{A0} \sim 0.18$–0.25 (H shifts a_0 by 18–25 per cent per σ)
- $\alpha_{A0}/\alpha_{\rm Vf} \approx 4{:}1$ (a_0 channel 4× stronger)
- 1D accessible fraction: ∼30 per cent; 2D recoverable: ∼83 per cent

8.4 Implications for MOND

If the a_0 variation driven by halo shape is confirmed at larger N, it would challenge interpretations of a_0 as a fundamental constant. The 4:1 asymmetry in the a_0 channel would be difficult to accommodate in standard MOND frameworks. However, we note that the apparent variation could reflect fitting artefacts induced by non-circular motions from triaxial potentials rather than genuine a_0 variation — a distinction that requires hydrodynamic simulations to resolve.

8.5 Implications for ΛCDM

The result is natural in ΛCDM. Triaxial haloes are a generic prediction of collisionless structure formation (Allgood et al. 2006; Vera-Ciro et al. 2011). The implied axis ratio distribution ($b/a \sim 0.65$–0.75, scatter 0.10–0.18) is consistent with N-body predictions. The regime dependence (activation at $V_{\rm flat} \gtrsim 120\,\mathrm{km\,s^{-1}}$) may reflect the mass-dependence of halo shapes observed in simulations.

8.6 Implications for the BTFR

The bilateral coupling implies that baryonic Tully–Fisher relation (BTFR) scatter is not random but contains systematic physical information about halo shape. If confirmed, part of the residual BTFR scatter would trace halo triaxiality rather than measurement error or stochastic galaxy-to-galaxy variation, with implications for using the BTFR as a distance indicator.

8.7 Remaining caveats

(i) **Sample size ($N = 7$).** All 2D statistics have wide confidence intervals. LOO is universally positive but NGC 2841 has moderate leverage (drop of 0.24 when removed). Definitive confirmation requires $N \geq 20$.

(ii) **Single-survey dependence.** All 2D data come from THINGS. Cross-survey replication is essential before the carrier identification can be considered definitive. We verified that the apparent MaNGA overlap reported in our census was a target-selection artefact: none of the SPARC quality-sample galaxies appear in the MaNGA observed catalogue (SDSS DR17 mangaDrpAll; $N_{\rm MaNGA} = 11{,}273$), because MaNGA targets lie predominantly at $z > 0.01$ while SPARC galaxies reside in the local volume ($z < 0.007$). Immediate expansion from existing public IFU surveys is therefore unavailable; dedicated HI/IFU follow-up observations are required.

(iii) **Triaxiality–assembly degeneracy.** Triaxiality and assembly history score 100 per cent versus 96 per cent in our hypothesis tests. In CDM cosmology, formation history drives triaxiality, making them observationally degenerate at present. Resolving this requires cosmological simulations with known triaxiality parameters.

(iv) **No simulation comparison.** We have not tested the predicted DQ–m=2 correlation against galaxies from IllustrisTNG or FIRE with known halo shapes. Such a comparison would provide independent confirmation or falsification.

9 CONCLUSIONS

Across nine analysis programs, we have pursued a single question: what is the physical nature of the hidden common-cause state H that drives the bilateral coupling between VfResid and a0Resid in SPARC rotation curves?

We do not claim final identification of the fundamental physics. We do claim the following:

(i) The hidden common-cause state H is real: per-galaxy a_0 varies systematically, driven by H through a bilateral coupling at $r = 0.77$ (LOO, $p < 0.001$, $N = 55$, replicated on $N = 59$ independent galaxies).

(ii) The information ceiling of 1D rotation curves is genuine: 88.5 per cent of H is inaccessible from rotation-curve features, regardless of representation (scalar, radial profile, or 71-dim map).

(iii) The outer-halo m=2 velocity-field mode is the strongest observational carrier of H identified to date (r = 0.85–0.91 across two independent pipelines, $p = 0.005$, 11/11 red team tests pass including 8/8 critical).

(iv) The leading physical interpretation is a triaxial or closely related halo-shape state, supported by cosmological Monte Carlo (concentration-only models fail; triaxiality required) and the outer-halo persistence of the signal.

(v) The information ceiling arises because azimuthal averaging destroys the m=2 angular structure by construction, preserving only integral mass shifts that produce the bilateral coupling.

The information was not in the rotation curve itself. It was in the angular structure of the velocity field that the rotation curve averages over.

The current 2D confirmation is strong but sample-limited ($N = 7$, THINGS only), and immediate public-survey expansion is unavailable because the apparent MaNGA overlap was a target-selection artefact rather than an observed-sample overlap. Definitive confirmation therefore requires dedicated HI or IFU follow-up of high-|DQ| SPARC galaxies — specifically ESO 563-G021 (DQ = +2.27), UGC 06787 (+1.98), and UGC 09037 (−1.78) — to reach $N \geq 20$ across independent facilities.

ACKNOWLEDGEMENTS

This analysis uses data from the SPARC database (Lelli et al. 2016) and THINGS survey (Walter et al. 2008). We thank the teams that made these datasets publicly available. This work was conducted as an independent computational analysis without institutional affiliation.

DATA AVAILABILITY

All analysis scripts (37 files), machine-readable results (34 files), and supplementary data are publicly available on Zenodo:

- Version 13 (this paper): DOI: 10.5281/zenodo.19444465
- Concept DOI: 10.5281/zenodo.19430633

The SPARC database is available at urlhttp://astroweb.cwru.edu/SPARC/. THINGS data are available at `https://www.mpia.de/THINGS/Data.html`.

REFERENCES

Allgood B., Flores R. A., Primack J. R., Kravtsov A. V., Wechsler R. H., Faltenbacher A., Bullock J. S., 2006, Monthly Notices of the Royal Astronomical Society, 367, 1781

Lelli F., McGaugh S. S., Schombert J. M., 2016, The Astronomical Journal, 152, 157

Lelli F., McGaugh S. S., Schombert J. M., Pawlowski M. S., 2017, The Astrophysical Journal, 836, 152

Marra V., Rodrigues D. C., de Oliveira A. S., 2020, Monthly Notices of the Royal Astronomical Society, 494, 2875

McGaugh S. S., Lelli F., Schombert J. M., 2016, Physical Review Letters, 117, 201101

Rodrigues D. C., Marra V., del Popolo A., Davari Z., 2018, Nature Astronomy, 2, 668

Vera-Ciro C. A., Sales L. V., Helmi A., Frenk C. S., Navarro J. F., Springel V., Vogelsberger M., White S. D. M., 2011, Monthly Notices of the Royal Astronomical Society, 416, 1377

Walter F., Brinks E., de Blok W. J. G., Bigiel F., Kennicutt R. C., Thornley M. D., Leroy A., 2008, The Astronomical Journal, 136, 2563

This paper has been typeset from a TEX/LATEX file prepared by the author.